Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Prove by direct deductive reasoning that the difference between the squares of any two odd integers is always divisible by 8.

1. Let m and n be any two odd integers. Then they can be expressed algebraically as $m = 2k + 1$ and $n = 2j + 1$ for some integers $k, j \in \mathbb{Z}$. 2. Form the difference of their squares: $m^2 - n^2 = (2k+1)^2 - (2j+1)^2 = (4k^2 + 4k + 1) - (4j^2 + 4j + 1) = 4k^2 + 4k - 4j^2 - 4j = 4[k(k+1) - j(j+1)]$. 3. Analyze the products $k(k+1)$ and $j(j+1)$: For any integer k, exactly one of k or $k+1$ is even. Therefore, the product of any two consecutive integers is always divisible by 2.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The sum of the first n terms of an arithmetic series is given by $S_n = 4n^2 - 3n$. (a) Find the first term u_1 and the second term u_2. (b) Find the common difference d. (c) Find an explicit expression for the nth term u_n, and calculate u_{25}.

(a) $u_1 = S_1 = 4(1)^2 - 3(1) = 1$. $S_2 = 4(2)^2 - 3(2) = 16 - 6 = 10$. $u_2 = S_2 - S_1 = 10 - 1 = 9$. (b) Common difference: $d = u_2 - u_1 = 9 - 1 = 8$. (c) General term: $u_n = u_1 + (n-1)d = 1 + (n-1)(8) = 8n - 7$. Alternatively: $u_n = S_n - S_{n-1} = (4n^2 - 3n) - [4(n-1)^2 - 3(n-1)] = 8n - 7$. For $n = 25$: $u_{25} = 8(25) - 7 = 200 - 7 = 193$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In triangle ABC, side $a = 7$ cm, side $b = 9$ cm, and angle $A = 42^\circ$. (a) Show that two distinct triangles can be formed with these given measurements. (b) Calculate both possible values for angle B and the corresponding values of side c.

(a) Height of triangle from vertex C: $h = b\sin A = 9\sin 42^\circ \approx 9(0.6691) = 6.022$ cm. Since $h < a < b$ (i.e. $6.022 < 7 < 9$), the side a is long enough to reach the base line on either side of the altitude. Therefore, exactly two non-congruent triangles exist. ■ (b) Sine rule: $\frac{\sin B}{b} = \frac{\sin A}{a} \Rightarrow \sin B = \frac{9\sin 42^\circ}{7} \approx \frac{6.0221}{7} \approx 0.8603$. First triangle (acute): $B_1 = \arcsin(0.8603) \approx 59.35^\circ \approx 59.4^\circ$..

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Three mobile antennas are located at coordinates $A(0,0)$, $B(6,0)$, and $C(2,4)$. (a) Find the equation of the perpendicular bisector of segment AB. (b) Find the equation of the perpendicular bisector of segment AC. (c) Find the coordinates of the Voronoi vertex (circumcentre) equidistant from all three antennas, and calculate the radius of the circumcircle.

(a) For segment AB: $A(0,0)$ and $B(6,0)$ lie on the x-axis. Midpoint $M_{AB} = (3,0)$. The perpendicular bisector is the vertical line $x = 3$. (b) For segment AC: $A(0,0)$ and $C(2,4)$. Midpoint $M_{AC} = \left(\frac{0+2}{2}, \frac{0+4}{2}\right) = (1,2)$. Gradient $m_{AC} = \frac{4-0}{2-0} = 2$. Perpendicular gradient $m_{\perp} = -\frac{1}{2}$. Equation: $y - 2 = -\frac{1}{2}(x-1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2} \Leftrightarrow x + 2y = 5$. (c) The Voronoi vertex is the intersectio.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Six wines were evaluated by two independent sommeliers, X and Y, giving the following ratings: Wine: A, B, C, D, E, F Judge X: 88, 72, 95, 60, 81, 75 Judge Y: 84, 78, 91, 65, 85, 70 (a) Construct a table of ranks for both judges. (b) Calculate Spearman's rank correlation coefficient r_s. (c) State what this coefficient indicates about the judges' agreement.

(a) Ranking in descending order (highest rating = rank 1): Judge X: 95 (1), 88 (2), 81 (3), 75 (4), 72 (5), 60 (6). Judge Y: 91 (1), 85 (2), 84 (3), 78 (4), 70 (5), 65 (6). Table of paired ranks and differences $d_i = R_X - R_Y$: Wine A: $R_X = 2, R_Y = 3 \Rightarrow d_1 = -1, d_1^2 = 1$ Wine B: $R_X = 5, R_Y = 4 \Rightarrow d_2 = 1, d_2^2 = 1$ Wine C: $R_X = 1, R_Y = 1 \Rightarrow d_3 = 0, d_3^2 = 0$ Wine D: $R_X = 6, R_Y = 6 \Rightarrow d_4 = 0, d_4^2 = 0$ Wine E: $R_X = 3, R_Y = 2 \Rightarrow d_5 = 1, d_5^2 = 1$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A small business borrows USD 40,000 to renovate its premises. The loan carries an annual nominal interest rate of 7.2% compounded monthly, with equal monthly repayments made at the end of each month over 6 years. (a) Calculate the monthly repayment amount. (b) Calculate the total amount repaid and the total interest charged on the loan. (c) After making payments for 3 full years, determine the outstanding loan balance.

(a) Total periods $N = 6 \times 12 = 72$. Monthly interest rate $r = \frac{7.2\%}{12} = 0.6\% = 0.006$. Using the present value of an annuity formula: $PV = PMT\left[\frac{1-(1+r)^{-N}}{r}\right] \Rightarrow 40,000 = PMT\left[\frac{1-(1.006)^{-72}}{0.006}\right]$. $(1.006)^{-72} \approx 0.65028 \Rightarrow 1 - 0.65028 = 0.34972$. Fraction: $\frac{0.34972}{0.006} \approx 58.2866$. $PMT = \frac{40,000}{58.2866} \approx USD686.26$. (b) Total amount repaid: $72 \times 686.264 = USD49,411.01$. Total intere.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A particle moves along a straight line such that its velocity in m/s at time t seconds is given by $v(t) = 6t - 2t^2$ for $0 \le t \le 4$. (a) Find the acceleration of the particle at $t = 2$ s. (b) Determine the net displacement of the particle between $t = 0$ and $t = 4$. (c) Calculate the total distance travelled by the particle over the interval $[0, 4]$.

(a) Acceleration: $a(t) = v'(t) = 6 - 4t$. At $t = 2$: $a(2) = 6 - 4(2) = -2$ m/s^2. (b) Net displacement is the integral of velocity: $\Delta s = \int_0^4 (6t - 2t^2)\,dt = \left[3t^2 - \frac{2}{3}t^3\right]_0^4 = (3(16) - \frac{2}{3}(64)) - 0 = 48 - \frac{128}{3} = \frac{16}{3} = 5\frac{1}{3}$ m. (c) Velocity vanishes when $6t - 2t^2 = 0 \Rightarrow 2t(3 - t) = 0 \Rightarrow t = 0$ and $t = 3$. For $0 \le t \le 3$, $v(t) \ge 0$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A curve has equation $f(x) = \frac{2x^2}{x^2 + k}$ for some positive constant $k > 0$. (a) Write down the horizontal asymptote of $y = f(x)$. (b) Find $f'(x)$ in terms of k. (c) Given that the gradient of the tangent to the curve at $x = 1$ is equal to $\frac{1}{2}$, find all exact possible values of k.

(a) $\lim_{x \to \pm\infty} \frac{2x^2}{x^2 + k} = \lim_{x \to \pm\infty} \frac{2}{1 + k/x^2} = 2$. Horizontal asymptote: $y = 2$. (b) Apply the quotient rule with $u = 2x^2$ and $v = x^2 + k$: $u' = 4x$, $v' = 2x$. $f'(x) = \frac{4x(x^2 + k) - 2x^2(2x)}{(x^2 + k)^2} = \frac{4x^3 + 4kx - 4x^3}{(x^2 + k)^2} = \frac{4kx}{(x^2 + k)^2}$. (c) At $x = 1$, $f'(1) = \frac{1}{2}$: $\frac{4k(1)}{(1^2 + k)^2} = \frac{1}{2} \Rightarrow \frac{4k}{(k + 1)^2} = \frac{1}{2} \Rightarrow 8k = (k + 1)^2$. Expand.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A closed cylindrical metal container with circular base of radius r cm and height h cm has a fixed volume of 250π cm^3. (a) Express h in terms of r. (b) Show that the total surface area is $A(r) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Find the radius r that minimizes the surface area, and show that for this optimal cylinder, the height equals the diameter ($h = 2r$).

(a) Volume of cylinder: $V = \pi r^2 h = 250\pi \Rightarrow h = \frac{250}{r^2}$. (b) Total surface area includes 2 circular ends plus curved surface: $A = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r\left(\frac{250}{r^2}\right) = 2\pi r^2 + \frac{500\pi}{r}$. (c) Differentiate with respect to r: $A'(r) = 4\pi r - \frac{500\pi}{r^2}$. Set $A'(r) = 0$: $4\pi r = \frac{500\pi}{r^2} \Rightarrow 4r^3 = 500 \Rightarrow r^3 = 125 \Rightarrow r = 5$ cm. Check second derivative.

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10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A continuous random variable X is normally distributed with mean μ and standard deviation σ. Given that $P(X < 45) = 0.12$ and $P(X > 85) = 0.08$, find the value of μ and σ correct to 3 significant figures.

1. Standardize using $Z = \frac{X-\mu}{\sigma}$: For the lower tail: $P(X < 45) = 0.12 \Rightarrow P(Z < z_1) = 0.12$. Using GDC `invNorm(0.12, 0, 1)`: $z_1 = -1.17499$. For the upper tail: $P(X > 85) = 0.08 \Rightarrow P(X < 85) = 0.92 \Rightarrow P(Z < z_2) = 0.92$. Using GDC `invNorm(0.92, 0, 1)`: $z_2 = 1.40507$. 2. Form the system of equations: $\frac{45-\mu}{\sigma} = -1.1750 \Rightarrow \mu - 1.1750\sigma = 45$. $\frac{85-\mu}{\sigma} = 1.4051 \Rightarrow \mu + 1.4051\sigma = 85$. 3. Subtract the first.

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the function $f(x) = x^2e^{-x}$. (a) Find $f'(x)$ and express it in factored form. (b) Hence, find the coordinates of all stationary points of f. (c) Use the second derivative test to classify each stationary point as a local maximum or local minimum.

(a) Apply the product rule with $u = x^2$ and $v = e^{-x}$: $u' = 2x$, $v' = -e^{-x}$. $f'(x) = 2xe^{-x} + x^2(-e^{-x}) = (2x - x^2)e^{-x} = x(2-x)e^{-x}$. (b) Set $f'(x) = 0$. Since $e^{-x} > 0$ for all x: $x(2-x) = 0 \Rightarrow x = 0 \quad \text{or} \quad x = 2$. Coordinates: At $x = 0$: $f(0) = 0^2e^0 = 0 \Rightarrow (0, 0)$. At $x = 2$: $f(2) = 2^2e^{-2} = 4e^{-2} = \frac{4}{e^2} \approx 0.5413 \Rightarrow \left(2, \frac{4}{e^2}\right)$. (c) Compute $f''(x)$ by differentiating $(2x - x^2)e^{-x}$.

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the exact area of the finite region enclosed between the parabola $y = 3x - x^2$ and the line $y = x - 3$.

1. Find intersection points: $3x - x^2 = x - 3 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1) = 0 \Rightarrow x = -1 \quad \text{and} \quad x = 3.$ 2. On the interval $[-1, 3]$, the parabola is concave down and lies above the line: At $x = 0$: $y_{\text{parabola}} = 0 > -3 = y_{\text{line}}$. 3. Set up the definite integral: $A = \int_{-1}^{3} [(3x - x^2) - (x - 3)]\,dx = \int_{-1}^{3} (3 + 2x - x^2)\,dx$. 4. Integrate.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

In triangle PQR, angle PQR = 90 degrees. S is a point on PR such that QS is perpendicular to PR. The length of PQ is 15 cm and the length of QR is 20 cm. Let the length of PS be x cm and the length of SR be y cm. Find the value of x.

First, we find the length of the hypotenuse PR using the Pythagorean theorem in triangle PQR: $PR^2 = PQ^2 + QR^2$. $PR^2 = 15^2 + 20^2 = 225 + 400 = 625$. So, $PR = \sqrt{625} = 25$ cm. Triangle PQR is similar to triangle PSQ (by AA similarity, as both share angle P and have a right angle). Therefore, the ratio of corresponding sides is equal: $\frac{PS}{PQ} = \frac{PQ}{PR}$. Substituting the given values: $\frac{x}{15} = \frac{15}{25}$. Solving for x: $x = 15 \times \frac{15}{25} = 15 \times \frac{3}{5} = 9$ cm..

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The lifespan X in hours of an electronic component is normally distributed with mean $\mu = 1200$ hours and standard deviation $\sigma = 150$ hours. (a) Find the probability that a randomly selected component lasts more than 1350 hours. (b) In a batch of 500 components, estimate how many will last between 1000 and 1400 hours. (c) The manufacturer guarantees the top 10% longest-lasting components. Determine the minimum lifespan required to receive this guarantee.

(a) Standardize: $Z = \frac{1350 - 1200}{150} = \frac{150}{150} = 1.0$. $P(X > 1350) = P(Z > 1.0) = 1 - \Phi(1.0) \approx 1 - 0.8413 = 0.1587$ (15.9%). (b) For $1000 \le X \le 1400$: $Z_1 = \frac{1000 - 1200}{150} = -\frac{200}{150} = -1.3333$. $Z_2 = \frac{1400 - 1200}{150} = \frac{200}{150} = 1.3333$. Using GDC `normalcdf(-1.3333, 1.3333, 0, 1)`: $P \approx 0.8176$. Expected number: $500 \times 0.8176 \approx 408.8 \approx 409$ components. (c) We require.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Number & Algebra

In a factory, 60% of components are manufactured by Machine A and 40% by Machine B. It is known that 3% of components produced by Machine A are defective, whereas 5% of components produced by Machine B are defective. (a) Find the probability that a randomly chosen component is defective. (b) Given that a randomly chosen component is defective, find the probability that it was produced by Machine B. (c) Determine whether the events "component produced by Machine B" and "component is defective" are independent.

(a) By the law of total probability: $P(D) = P(A)P(D \mid A) + P(B)P(D \mid B) = (0.60)(0.03) + (0.40)(0.05)$ $= 0.018 + 0.020 = 0.038$ (3.8%). (b) By Bayes' Theorem / conditional probability: $P(B \mid D) = \frac{P(B \cap D)}{P(D)} = \frac{P(B)P(D \mid B)}{P(D)} = \frac{(0.40)(0.05)}{0.038} = \frac{0.020}{0.038} = \frac{20}{38} = \frac{10}{19} \approx 0.5263$ (52.6%). (c) Check the independence condition $P(B \cap D) = P(B)P(D)$: $P(B \cap D) = 0.020$. $P(B) \times P(D) = 0.40 \times 0.038 = 0.0152$..

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

In a 2D Cartesian plane, points A, B, C have coordinates $A(2, -1)$, $B(5, 3)$, and $C(-1, k)$. (a) Find the vectors $\vec{AB}$ and $\vec{AC}$ in component form. (b) Find the value of k for which $\vec{AB}$ is perpendicular to $\vec{AC}$. (c) For $k = 3$, calculate the exact cosine of the angle between $\vec{AB}$ and $\vec{AC}$.

(a) $\vec{AB} = (5 - 2; 3 - (-1)) = (3; 4)$. $\vec{AC} = (-1 - 2; k - (-1)) = (-3; k + 1)$. (b) Perpendicularity condition $\vec{AB} \cdot \vec{AC} = 0$: $3(-3) + 4(k + 1) = 0 \Rightarrow -9 + 4k + 4 = 0 \Rightarrow 4k - 5 = 0 \Rightarrow k = \frac{5}{4} = 1.25$. (c) When $k = 3$: $\vec{AC} = (-3; 4)$. Lengths: $|\vec{AB}| = \sqrt{3^2 + 4^2} = 5$, $|\vec{AC}| = \sqrt{(-3)^2 + 4^2} = 5$. Dot product: $\vec{AB} \cdot \vec{AC} = 3(-3) + 4(4) = -9 + 16 = 7$..

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17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

Consider the binomial expansion of $\left(x^2 - \frac{2}{x}\right)^9$. (a) Write down the general term in terms of r. (b) Find the coefficient of x^3. (c) Find the term independent of x.

(a) General term: $T_{r+1} = \binom{9}{r}(x^2)^{9-r}\left(-\frac{2}{x}\right)^r = \binom{9}{r}(-2)^r x^{2(9-r)}x^{-r} = \binom{9}{r}(-2)^r x^{18-3r}$. (b) To find the coefficient of x^3, set the power of x equal to 3: $18 - 3r = 3 \Rightarrow 3r = 15 \Rightarrow r = 5$. Substitute $r = 5$: $T_6 = \binom{9}{5}(-2)^5 x^3 = 126 \times (-32)x^3 = -4032x^3$. The coefficient of x^3 is -4032. (c) The term independent of x has power 0: $18 - 3r = 0 \Rightarrow 3r = 18 \Rightarrow r = 6$. Substitute.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Calculus

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let X be the number of correct answers. (a) State the distribution of X including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of X.

(a) $X \sim B(n, p)$ with $n = 20$ and $p = 0.25$. (b) $P(X \geq 10) = 1 - P(X \leq 9)$. Using GDC `binomcdf(20, 0.25, 9)`: $P(X \leq 9) \approx 0.9861$. $P(X \geq 10) = 1 - 0.9861 = 0.0139 \quad (1.39\%)$. (c) Mean: $\mu = np = 20 \times 0.25 = 5$. Standard deviation: $\sigma = \sqrt{np(1-p)} = \sqrt{20 \times 0.25 \times 0.75} = \sqrt{3.75} \approx 1.936 \approx 1.94$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 AI SL — Specific Topics

Practice 2.1: The function $f(x) = 3 - \ln(2x + 1)$ is defined for $x > -0.5$. Find an expression for $f^{-1}(x)$, state its domain and range, and write down the coordinates of the intercept of $y = f^{-1}(x)$ with the y-axis.

1. Let $y = 3 - \ln(2x + 1)$. 2. Rearrange: $\ln(2x + 1) = 3 - y \Rightarrow 2x + 1 = e^{3-y} \Rightarrow x = \frac{e^{3-y} - 1}{2}$. 3. Swap variables: $f^{-1}(x) = \frac{e^{3-x} - 1}{2}$. 4. Domain of f^{-1} is the range of f, which is $\mathbb{R}$ (all real numbers). Range of f^{-1} is the domain of f, $(-0.5, \infty)$. 5. The y-intercept occurs at $x = 0$: $f^{-1}(0) = \frac{e^3 - 1}{2}$. Coordinates: $\left(0, \frac{e^3 - 1}{2}\right)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number & Algebra

Practice 4.2: Bivariate data for study hours x and exam mark y across 10 students yield $\bar{x} = 14.5$, $\bar{y} = 72.0$, and Pearson's correlation coefficient $r = 0.88$. The regression line of y on x has equation $y = 2.4x + c$. (a) Find the value of c. (b) Interpret the slope 2.4 in context. (c) Estimate the exam mark of a student who studies 18 hours.

(a) The regression line must pass through the mean point $(\bar{x}, \bar{y}) = (14.5, 72.0)$: $72.0 = 2.4(14.5) + c \Rightarrow 72.0 = 34.8 + c \Rightarrow c = 72.0 - 34.8 = 37.2$. Thus $y = 2.4x + 37.2$. (b) The slope 2.4 indicates that for each additional hour of study, a student's predicted exam mark increases by 2.4 marks. (c) For $x = 18$: $\hat{y} = 2.4(18) + 37.2 = 43.2 + 37.2 = 80.4$ marks.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 6$ in cubic functions, evaluate $f(2)$.

a. 10 ✓

b. 12

c. 4

d. 14

Substitute $x = 2$: $f(2) = 2(2) + 6 = 10$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in power functions, evaluate $f(2)$.

a. 13 ✓

b. 16

c. 6

d. 17

Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in direct variation, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cubic functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for power functions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for hyperbolas and inverse variation consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 9$ in hyperbolas and inverse variation, evaluate $f(4)$.

a. 29 ✓

b. 34

c. 20

d. 33

Substitute $x = 4$: $f(4) = 5(4) + 9 = 29$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for transformations of hyperbolas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 3$ in transformations of hyperbolas, evaluate $f(4)$.

a. 27 ✓

b. 33

c. 24

d. 31

Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 4$ in graphing square root functions, evaluate $f(3)$.

a. 10 ✓

b. 12

c. 6

d. 14

Substitute $x = 3$: $f(3) = 2(3) + 4 = 10$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x) = \frac{1}{x-1}$ and the function $g(x) = x^2$. Let $h(x) = f(g(x))$. The graph of $y = h(x)$ is shown below. Determine the equation of the vertical asymptote(s) of $y = h(x)$.

a. x = 0

b. x = 1

c. **x = -1** ✓

d. x = 0 and x = 1

The function $h(x)$ is a composite function, where $h(x) = f(g(x))$. We are given $f(x) = \frac{1}{x-1}$ and $g(x) = x^2$. Substituting $g(x)$ into $f(x)$, we get $h(x) = f(x^2) = \frac{1}{x^2-1}$. To find the vertical asymptotes of $y = h(x)$, we need to find the values of x for which the denominator is zero and the numerator is non-zero. The denominator $x^2 - 1 = 0$ when $x^2 = 1$, which means $x = 1$ or $x = -1$. The numerator is a constant 1, which is never zero. Therefore, the vertical asymptotes are at.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.

a. 3

b. 4

c. **5** ✓

d. 6

To find the intersection points of the two functions, we set $f(x) = g(x)$. $x^2 - 6x + 11 = 2x - 1$ Rearrange the equation to form a quadratic equation: $x^2 - 6x - 2x + 11 + 1 = 0$ $x^2 - 8x + 12 = 0$ We can factor this quadratic equation. We are looking for two numbers that multiply to 12 and add to -8. These numbers are -6 and -2. $(x - 2)(x - 6) = 0$ This gives us two possible solutions for x: $x = 2$ or $x = 6$. We are given that one intersection point has an x-coordinate of 2. Therefore, the x-coordinate of.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(2, -3)$. Determine the value of k.

a. $k = 5$ ✓

b. $k = -5$

c. $k = 1$

d. $k = -1$

The function is given by $f(x) = x^2 - 6x + k$. The graph of the function passes through the point $(2, -3)$. This means that when $x = 2$, $f(x) = -3$. We can substitute these values into the function equation: $-3 = (2)^2 - 6(2) + k$ $-3 = 4 - 12 + k$ $-3 = -8 + k$ To solve for k, add 8 to both sides of the equation: $-3 + 8 = k$ $5 = k$ Therefore, the value of k is 5.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown below. The graph passes through the points $(1, 0)$, $(5, 0)$, and $(0, -10)$. Find the value of a.

a. -2 ✓

b. -1

c. 1

d. 2

The quadratic function has roots at $x = 1$ and $x = 5$. Therefore, it can be written in the factored form $f(x) = a(x - 1)(x - 5)$. We are given that the graph passes through the point $(0, -10)$. Substituting these coordinates into the factored form: $-10 = a(0 - 1)(0 - 5)$ $-10 = a(-1)(-5)$ $-10 = 5a$ Dividing both sides by 5, we get $a = -2$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

a. **(1, 0)** ✓

b. (3, 0)

c. (0, 3)

d. (0, 1)

To find the x-intercepts of the function $f(x) = x^2 - 4x + 3$, we set $f(x) = 0$. This gives us the quadratic equation $x^2 - 4x + 3 = 0$. We can factor this equation as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. Therefore, the x-intercepts are at $(1, 0)$ and $(3, 0)$. The point with the smaller x-value is $(1, 0)$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y -intercept.

a. **6** ✓

b. -3

c. 3

d. -6

The y-intercept of a function is the value of y when $x = 0$. For the function $f(x) = -2x + 6$, we substitute $x = 0$ into the equation: $f(0) = -2(0) + 6$ $f(0) = 0 + 6$ $f(0) = 6$ Therefore, the y-intercept is 6.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept?

a. $(0, -3)$ ✓

b. $(0, 3)$

c. $(-1, 0)$

d. $(3, 0)$

The y-intercept of a function occurs when $x = 0$. Substituting $x = 0$ into the function $f(x) = -x^2 + 4x - 3$ gives $f(0) = -(0)^2 + 4(0) - 3 = -3$. Therefore, the coordinates of the y-intercept are $(0, -3)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Calculus

Consider the two functions $f(x) = ax^2 + bx + c$ and $g(x) = dx + e$. The graph of $f(x)$ is a parabola with its vertex at $(3, -1)$ and passes through the point $(1, 3)$. The graph of $g(x)$ is a straight line passing through the points $(2, 5)$ and $(6, -3)$. If the graphs of $f(x)$ and $g(x)$ intersect at two points, and the x-coordinate of one of these intersection points is p, find the value of $\frac{a+d}{p}$ given that p is an integer.

a. 3

b. **2 ✓**

c. -2

d. -3

First, we find the equation for $f(x)$. Since the vertex is at $(3, -1)$, we can write $f(x) = a(x-3)^2 - 1$. Given that the graph passes through $(1, 3)$: $3 = a(1-3)^2 - 1$ $3 = a(-2)^2 - 1$ $3 = 4a - 1$ $4 = 4a$ $a = 1$. So, $f(x) = 1(x-3)^2 - 1 = x^2 - 6x + 9 - 1 = x^2 - 6x + 8$. Thus, $a = 1$. Next, we find the equation for $g(x)$. The slope d is $\frac{3-2}{1-0} = \frac{1}{1} = 1$. Using the point-slope form with $(0, 2)$: $y - 2 = 1(x - 0)$ $y = x + 2$. So, $g(x) = x + 2$. Thus, $d = 1$. Now, we find t.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 AI SL — Specific Topics

Consider the functions $f(x) = |x - a|$ and $g(x) = b$, where a and b are constants. The graph of $y = f(x)$ is a V-shaped graph with its vertex at $(a, 0)$. The graph of $y = g(x)$ is a horizontal line. If the graphs of $y = f(x)$ and $y = g(x)$ intersect at exactly one point, what is the relationship between a and b for the graphs to intersect at a single point?

a. a = b

b. a = -b

c. **b = 0 ✓**

d. a = 0

The function $f(x) = |x - a|$ represents an absolute value function whose graph is a V-shape with its vertex at the point $(a, 0)$. The function $g(x) = b$ represents a horizontal line at the height $y = b$. For the graphs of $y = f(x)$ and $y = g(x)$ to intersect at exactly one point, the horizontal line $y = b$ must pass through the vertex of the V-shaped graph $y = |x - a|$. The vertex of $y = |x - a|$ is at $(a, 0)$. Therefore, for the intersection to occur at a single point, the y-coordinate of the horizontal line must be equal.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Number & Algebra

Consider the function $f(x) = 2x - 3$. Find the value of $f(3)$.

a. 3

b. **6** ✓

c. 9

d. 0

To find the value of $f(3)$, we substitute $x = 3$ into the function definition $f(x) = 2x - 3$. \\ \\ $f(3) = 2(3) - 3$ \\ $f(3) = 6 - 3$\\ $f(3) = 3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Geometry & Trigonometry

The point $(0, -3)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x - 1) - 4$.

a. $(1, -7)$ ✓

b. $(-4, -2)$

c. $(1, 1)$

d. $(-1, -7)$

Working through this functions step: Horizontal shift by 1 and vertical shift by -4: $(x, y) \to (x + 1, y - 4)$, giving $(1, -7)$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Statistics & Probability

Solve $2x^2 - 22x + 56 = 0$. Find the larger root.

a. 4

b. **7** ✓

c. 2

d. 3

Working through this functions step: $x = \frac{-(-22) \pm \sqrt{36}}{2(2)} \Rightarrow x = 4$ or $x = 7$. Larger root: 7.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Calculus

Let $f(x) = 3x^2 - 7x$. Find the equation of the axis of symmetry.

a. $x = -\frac{7}{3}$

b. $x = \frac{7}{3}$

c. $x = -\frac{7}{6}$

d. $x = \frac{7}{6}$ ✓

Working through this functions step: Axis of symmetry: $x = -\frac{b}{2a} = -\frac{-7}{2(3)} = \frac{7}{6}$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 AI SL — Specific Topics

Convert 210° to radians, giving your answer as an exact multiple of π.

a. $\frac{5\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{7\pi}{6}$ ✓

d. $\frac{7\pi}{4}$

Working through this trigonometry step: $210^\circ \times \frac{\pi}{180} = \frac{7\pi}{6}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number & Algebra

Let $f(x) = 2x^2 + 6x - 5$. Find the y-intercept of the graph of f.

a. -15

b. -10

c. **-5** ✓

d. 5

Working through this functions step: The y-intercept occurs at $x = 0$: $f(0) = -5$.

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